

Experiment No: 03

Date : _____

Roll No: _____

Aim : Draw basic 2D shapes like: line, rectangle circle, ellipse, arc polygon using computer aided drafting approach

Theory:

2-D shapes:

2D shapes can be defined as the plane figures that can be drawn on a flat (or plane) surface or a piece of paper. All the 2d shapes have various parameters such as area and perimeter. Some of the 2d shapes contain sides and corners, whereas some have curved boundaries.

Types of 2D Shapes:

1. Circle:

A circle is a closed 2d figure in which the set of all the points in the plane is equidistant from a given point called “center”. The distance from the center to the outer line of the circle is called a radius. The example of the circle in real life is wheels, pizzas, orbit, etc.

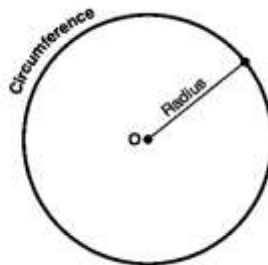


Fig : Circle

2. Line:

A line is made of a set of points which is extended in opposite directions infinitely.

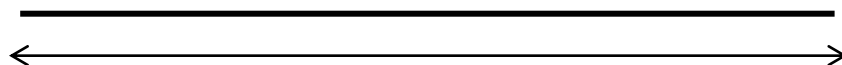


Fig : Line

3. Rectangle:

A rectangle is a 2d shape which has four sides, where the opposite sides are equal and parallel to each other. All the angles of a rectangle are equal to 90° . A brick, TV, cardboard, which has length and breadth are examples of the rectangle.



Fig : Rectangle

4. Ellipse :

An ellipse is a curve formed by a plane, that intersects a cone at an angle with respect to the base. It is a curve surrounded by two focal points. Ellipse is a locus of points, the sum of whose distances from two foci is constant.

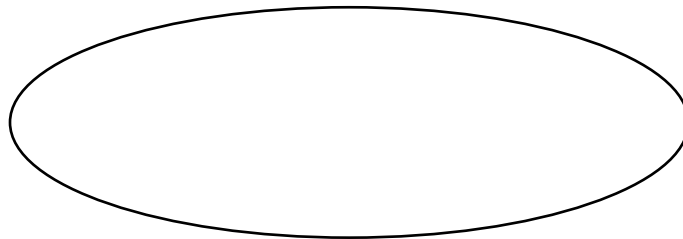


Fig : Ellipse

5. Polygon :

A polygon is a closed figure where the sides are all line segments. Each side must intersect exactly two other sides but only at their endpoints. The sides must be noncollinear and have a common endpoint.

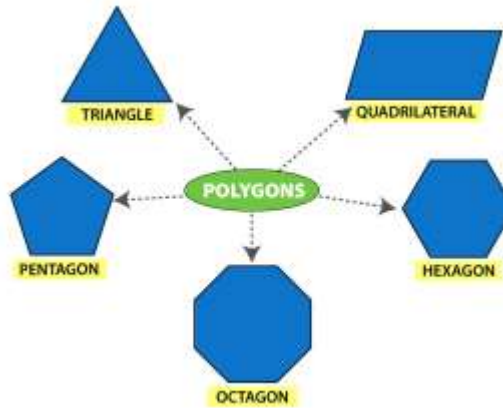


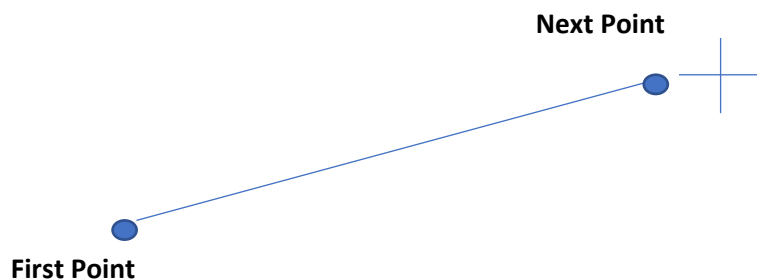
Fig : Polygon

Auto – CAD :

AutoCAD is the original CAD software used by millions around the world. It can be used to create precise 2D and 3D drawings and models, as well as electrical diagrams, construction drawings, and more.

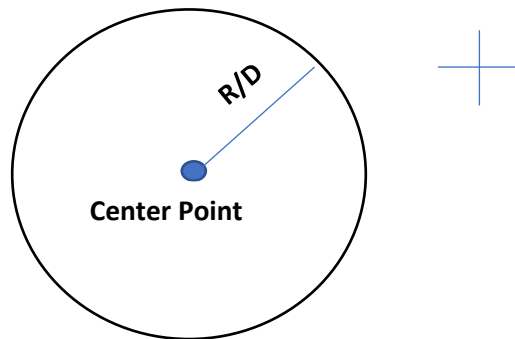
1. Line Command :

1. Open the AutoCAD software.
2. Select the Line icon from the ribbon panel or type L or Line and press Enter on the command line. ...
3. Specify the starting point and endpoint using the cursor on the workspace or drawing area.
4. Press Enter or Esc to exit.



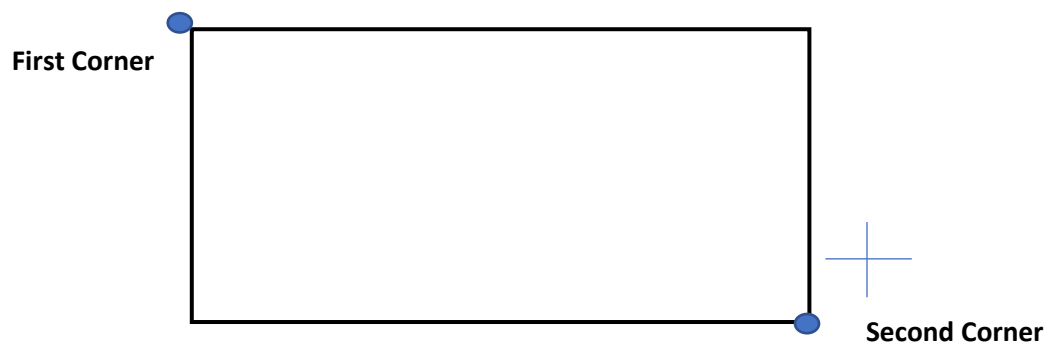
2. Circle Command :

1. Open the AutoCAD software.
2. Select the Circle icon from the ribbon panel or type C or Line and press Enter on the command line. ...
3. Specify the Center point of the circle using the cursor on the workspace or drawing area
4. Enter radius or diameter of the circle .
5. Press Enter to exit.



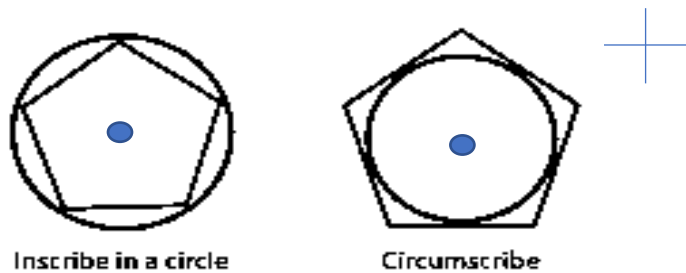
3. Rectangle Command :

1. Click Home tab > Draw panel > Rectangle. Find.
2. Specify the first corner of the rectangle.
3. Enter A for Area.
4. Enter the area.
5. Do one of the following: Enter L to enter the length. The width is calculated based on the length and the area. Enter W to enter the width.
6. Enter the value for the length or width



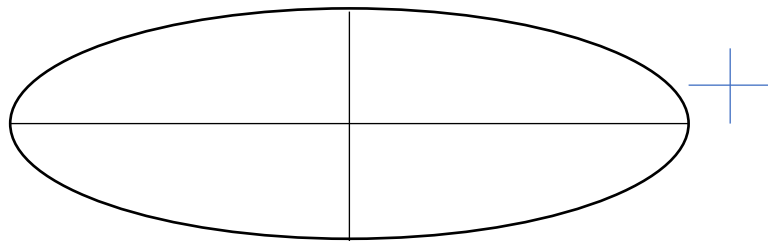
4. Polygon Command :

1. Click Home tab > Draw panel > Rectangle. Or type POL on the command line and press Enter.
2. Specify the number of sides.
3. Select method : Inscribe or Circumscribe
4. Specify the center of polygon on the viewport or workspace
5. Press Enter



5. Ellipse :

1. Click Home tab Draw panel Ellipse drop-down Elliptical Arc.
2. Specify endpoints for the first axis
3. Specify a distance to define half the length of the second axis
4. Specify the start angle
5. Specify the end angle



Conclusion :